Manuscript cover letter 26 October 2026

Dear Editors of The Electronic Journal of Combinatorics,

Please consider *The minimum number of distinct abelian squares in a binary circular word*, by Deep Bhattacharjee.

The paper proves the sharp lower bound $\lfloor (n-1)/2 \rfloor$ for every binary circular word of positive length and classifies all equality cases. It proves the odd-length part of Simpson's Conjecture 0.6 and corrects the even-length list by adding the alternating words when $n \equiv 2 \pmod 4$, $n \geq 6$. The argument is self-contained and proceeds through a double count of square witnesses and a classification of rigid words with a tight letter. The six principal conclusions are stated and proved in the accompanying Lean development.

An earlier preprint is available at `https://hexagonmath.org/2610.00473v1`. This edition uses the amsart layout and retains the proof. Code and verification information are available from `https://github.com/creelie/literate-octo-lamp`.

The use of Claude in the original research and of OpenAI Codex in this edition's audit and preparation is disclosed in the manuscript and accompanying provenance. Funding and competing interests are stated in the manuscript.

Yours sincerely,

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Prepared submission draft. Before sending, the author must confirm the journal's human-proof-understanding, exclusivity and recent-submission requirements listed in the accompanying submission notes. This letter has not been sent.